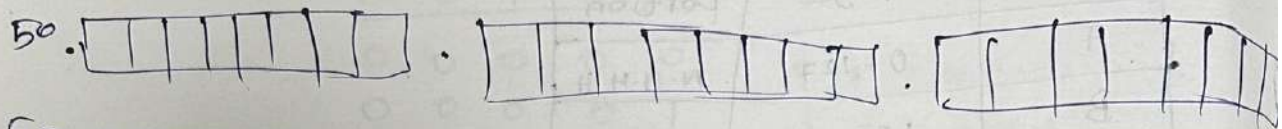


Day -4 - Parameters

Finding host IP in per Network

50.x.x.x → 50.0.0.0 → Network ID

50.255.255.255 → Broadcast IP



50.00000000 . 00000000 . 00000000

50.11111111 . 11111111 . 11111111

50.0.0.0 / 8 → Prefix Length = Number of Network

50.0.0.1 → First IP

50.0.0.2

↓
50.0.0.255

50.0.1.0

50.0.1.1

↓
50.0.1.255

50.0.2.0

50.0.2.1

↓ ↓
50.0.255.255

50.1.0.0

50.255.255.254 → Last IP

50.255.255.255

$$2^{24} = 2^{\text{host bit}}$$

$$2^{24} = 16777216$$

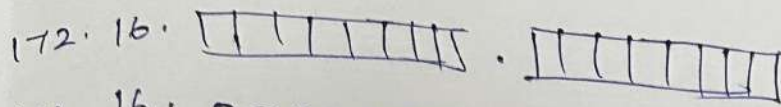
- 2 → Network ID and Broadcast IP

16777214 → total IP

Network ID - Permissions

Broadcast IP - Data

$$\begin{array}{r} 172 \cdot 16 \cdot 255 \cdot 117 \\ \hline N \cdot N \cdot H \cdot H \end{array}$$



$$172 \cdot 16 \cdot 00000000 \cdot 00000000$$



$$172 \cdot 16 \cdot 11111111 \cdot 11111111$$

$$172 \cdot 16 \cdot 0 \cdot 0 \mid 16$$

$$172 \cdot 16 \cdot 255 \cdot 255$$

$$\begin{array}{r} 192 \cdot 192 \cdot 81 \cdot 116 \\ \hline N \cdot N \cdot N \cdot H \end{array}$$

$$192 \cdot 192 \cdot 81 \cdot \boxed{}$$

$$0 \ 00000000$$

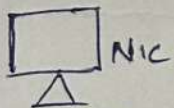
$$1 \ 11111111$$

$$192 \cdot 192 \cdot 81 \cdot 0 \mid 24$$

$$192 \cdot 192 \cdot 81 \cdot 1$$

$$192 \cdot 192 \cdot 81 \cdot 255$$

$$\begin{array}{r} 2^8 = 256 \\ - 2 \\ \hline 254 \end{array}$$



$$50 \cdot 0 \cdot 0 \cdot 10$$

Subnet mask :-

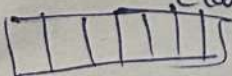
It is use to mention N/w Portion in IP

Condition

Put binary combination as 1 for N/w bit

$$\boxed{50} \cdot 0 \cdot 0 \cdot 10$$

Class A



$$255 \cdot 0 \cdot 0 \cdot 0$$

$$\boxed{172} \cdot \boxed{16} \cdot 0 \cdot 10$$

Class B

$$255 \cdot 255 \cdot 0 \cdot 0$$

$$192 \cdot 168 \cdot 10 \cdot 252$$

Class C

$$255 \cdot 255 \cdot 255 \cdot 0$$